Mineral Systems–Guided Prospectivity Screening with Spatial-Statistical Testing and Basement-Exposure Controls in the Kuldjuktau Range, Uzbekistan

Abdulla Baratov[1], Bum Han Lee[1,2*]

[1]Ore Geology Research Center, Korea Institute of Geoscience and Mineral Resources [2]Major of Geological Science, University of Science and Technology (UST) *leebh@kigam.re.kr

Abstract

This study evaluates a mineral-systems-guided approach to prospectivity screening in the Kuldjuktau Range of Central Kyzylkum, Uzbekistan, using open-access remote-sensing and archival data. Known mineral occurrences were not used to construct the evidence layers, but were used for spatial screening and model evaluation. The study area covers 2,365 km². Paleozoic basement is exposed over only about 21% of this area, yet contains 96% of the 310 archive-recorded mineral occurrences. Basement exposure alone therefore concentrates known occurrences about 4.6-fold, providing a simple regional baseline against which evidence-based rankings can be evaluated.

Twenty-two candidate evidence layers were assembled as proxies for geological controls or surface expressions relevant to mineral systems, derived from ASTER, Sentinel-1/2, EMIT, Landsat-8 TIRS, multi-source DEMs, archival geochemistry, and mapped geological structures. Each layer was percentile-ranked, and the layers were combined by geometric mean, so that a location scores highly where multiple lines of evidence converge. Performance was measured on a 100 m grid as enrichment, defined as the occurrence density within the highest-ranked area relative to the overall occurrence density. Significance was assessed against a toroidal-shift null model, which relocates the whole occurrence set while preserving its internal spatial structure. Layers that failed this screening, or that did not improve convergence, were excluded; no spectral alteration index was retained.

Among the tested combinations, the strongest regional enrichment came from lineament density combined with a seasonally persistent land-surface temperature-gradient anomaly, derived from Landsat-8 TIRS observations over four seasons between December 2022 and November 2023. This indicator represents the lateral gradient of land-surface temperature, not a vertical subsurface geothermal gradient.

The combination captured 73% of known occurrences within the highest-ranked 20% of the area, a 3.66-fold enrichment (spatial-null p ≈ 0.005). When the same test was restricted to exposed basement, however, the enrichment did not persist (0.62-fold, p = 0.40). Much of the apparent regional signal therefore reflects the contrast between exposed basement and younger sedimentary cover, rather than prospectivity differences within the basement itself.

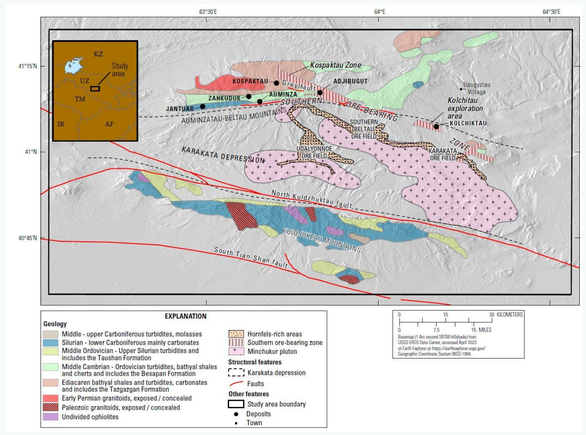
Retested within the basement, proximity to mapped quartz veins showed nominal enrichment (2.75-fold, p = 0.040), largely driven by gold occurrences, which constitute 68% of the basement set and are consistent with orogenic gold-quartz mineralization. An exploratory commodity-based comparison further suggests differing geological controls: gold occurrences lie nearer to mapped quartz veins than to intrusive bodies in 63% of cases, whereas rare-metal occurrences show the reverse in 63% (Fisher's exact p = 0.012; p = 0.009 in a separately digitized occurrence dataset). This contrast is consistent with a stronger magmatic association for the rare-metal occurrences. Given the number of layers tested and the small rare-metal sample, these within-basement associations should be regarded as exploratory. Using the adopted screening threshold and minimum-area filtering, the convergence surface delineated 67 zones covering 556 km² (23.5% of the domain); these are screening products, not validated exploration targets. Future work should quantify the predictive value of the selected layers beyond basement exposure, test robustness with independent or spatially separated validation, interpret the temperature-gradient signal geologically, and extend commodity-specific analyses where data permit.

GEOLOGY & DEPOSITS

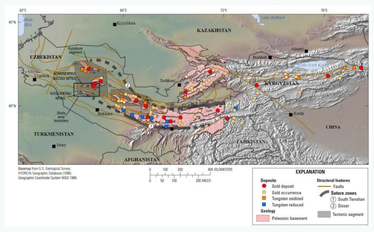
1 · Question and geological setting

Question: do open-data evidence layers rank prospectivity beyond what basement exposure alone explains?

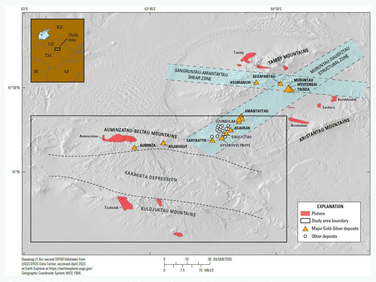
Kuldjuktau lies in Central Kyzylkum, ~100 km SW of the Muruntau deposit. Geologically it belongs to the Kyzylkum segment of the South Tien Shan fold-and-thrust belt (Central Asian Orogenic Belt) and is one of the windows where the Paleozoic basement is exposed, surrounded by the Mesozoic–Cenozoic sedimentary cover of the Turan Plate. The mapped domain covers 2,365 km².



Geology of the Kuldjuktau and Auminzatau ranges: Paleozoic granitoids, North Kuldjuktau and South Tien Shan faults, Karakata depression. Coyan et al. (2024), Fig. 3 (CC BY 4.0).



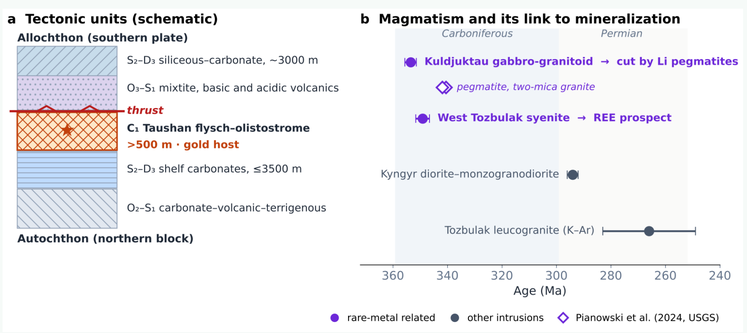
Position in the CAOB; South Tien Shan suture (Coyan et al., 2024, Fig. 1).



Central Kyzylkum gold province: Muruntau, Daugyztau, Amantaytau (Coyan et al., 2024, Fig. 2).

2 · Geological framework and magmatism

Gold is hosted by the sub-thrust Taushan mixtite; rare metals are linked to Early Carboniferous intrusions.



Structure and ages: Mirkamalov et al. (2025; U–Pb zircon, SHRIMP, K–Ar). ◇ Pianowski et al. (2024, USGS data release) — not yet reconciled with the table ages.

3 · Deposit types and genesis

Two main systems: structurally controlled orogenic gold and granite-related rare metals.

Type	Example	Genesis	Control	Role here
Orogenic gold (gold–quartz)	Taushan, Yangikazgan	CO_2-rich fluids → quartz veins (3–15 km)	Thrust suture, Taushan mixtite	Reference system
Rare-metal (Li) pegmatite	Aktosty–Shaydaraz	Fractionated granite → pegmatite; lepidolite (Li_2O 1.95 %, one sample), spodumene	Pegmatite–aplite veins cutting gabbro	Target
REE prospect	Tozbulak intrusion	Syenite, albitization; ΣREE 600–800 ppm (indication, not a resource)	Intrusion margins	Not tested
Graphite	Taskazgan deposit	Carbon metasomatism in mélange (debated)	Gabbro–carbonate contact	Not tested

Groves et al. (1998); Coyan et al. (2024); Mirkamalov et al. (2025)

4 · The occurrence set

310 archive-recorded occurrences

203 gold

27 rare metals (17 REE-bearing)

80 Hg, Cu, Zn, Al, other

- Gold is 68 % of the basement set, so it serves as the reference system; the rare-metal sample is small.
- Occurrences were used only to test the layers, never to build them. The catalogue is closed data: locations are not shown.

How the catalogue enters the analysis

Step (panel)	Occurrences used	What is reported
Exposure control (7)	309 in domain: 297 basement, 12 cover	Share by unit (96 % on 21 % of area)
Regional and basement tests (10)	309 / 297	Capture in top 20 %, enrichment, null p
Two systems (11)	Gold 202, rare metals 27 (basement)	Distances to veins / intrusions; Fisher test
Screening zones (9)	Not used to draw the zones	Zone areas only

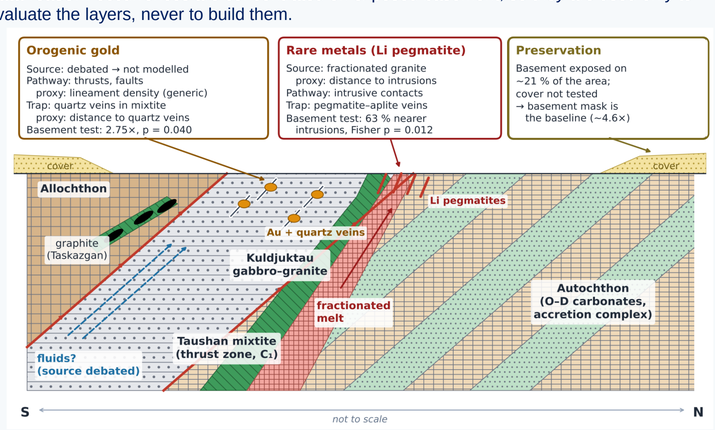
Source: Ministry of Mining Industry and Geology of Uzbekistan, occurrence catalogue compiled from State Geological Fund reports (unpublished, 2022; also used by Coyan et al., 2024).

MINERAL SYSTEMS → TEST

5 · Why a mineral-systems approach?

Geological reason: deposit models are well established (orogenic gold; LCT pegmatite; McCuaig et al., 2010), so each process is tied to a mapped footprint before occurrences are consulted; a locally debated process (the gold source) is not modelled.

Data reason: occurrences are concentrated on exposed basement, so they are used only to evaluate the layers, never to build them.



Schematic N–S section; geometry simplified and modified after Mirkamalov et al. (2025, Fig. 4). Process call-outs: this study.

Role of remote sensing: it maps the system's proxies — pathways (lineaments from DEMs and SAR), traps and footprints (alteration and lithology from ASTER, EMIT, TIR) and preservation (basement vs cover). Because it maps preservation so well, exposure is required as the baseline (panels 7, 10).

6 · Process → evidence → proxy → result

Process	Granite-related rare metals (target)	Orogenic gold (reference system)
Evidence in the range	Lepidolite–spodumene pegmatites cut the 353 Ma gabbro-granitoid; Li_2O 1.95 % in a lepidolite sample (Aktosty)	Au occurrences in sub-thrust Taushan mixtite
Source	Fractionated granitic magma — working hypothesis (Kuldjuktau, Tozbulak complexes)	Debated (metamorphic vs intrusion-related) → not modelled
Pathway	Intrusive contacts; pegmatite–aplite veins	Thrust suture zone, faults
Trap	Pegmatite / aplite bodies; albitized granite	Quartz veins in Taushan mixtite
Preservation	Basement window exposed on ~21 % of the area; cover not tested → basement mask = baseline (~4.6×)	
Proxy tested	Distance to intrusive bodies	Distance to mapped quartz veins
Result in basement	**63 % nearer intrusions; Fisher p = 0.012 (0.009)**	**2.75×, p = 0.040 (exploratory)**

Generic pathway / footprint proxies (lineament density, land-surface temperature gradient, spectral indices) were tested for both systems → panel 10. Source: Groves et al. (1998); Mirkamalov et al. (2025).

7 · The exposure problem

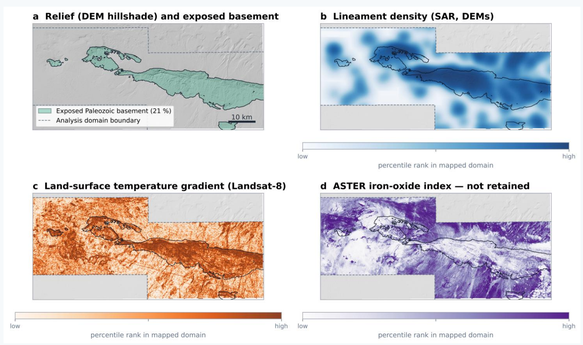
Basement is exposed on a fifth of the area but holds almost all known occurrences — the baseline any evidence layer must beat.

21 % of the area is exposed basement

96 % of 310 occurrences lie on it

~4.6× concentration from exposure alone

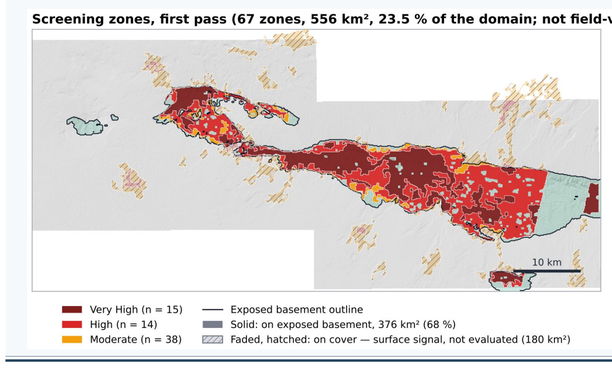
8 · Remote-sensing evidence and test design



- Base layers: lineament density for fluid pathways (multi-source DEM fusion incl. ALOS PALSAR, plus RADARSAT-1 SAR) × four-season Landsat-8 TIRS temperature gradient for lithological contrast.
- 22 candidates tested against them: ASTER band-ratio and TIR indices (5) and EMIT mineral maps (4) for alteration; archival geochemical halos (2); magnetics (1); geological map (5) and mapped structures (5) for pathways and traps. Sentinel-1/2 were tested the same way and not retained.
- Percentile rank → geometric mean; enrichment in the top 20 % on a 100 m grid vs a toroidal-shift null (500 shifts), repeated inside exposed basement.
- No alteration, geochemical or magnetic layer passed; 8 map- and structure-based layers passed over the whole area, none inside the basement.

What remote sensing gave: it maps the exposed basement and its structural framework well (a–c). Inside the basement no generic layer — satellite or map-based — separated the archive occurrences; only commodity-specific mapped controls did (panel 11). This is a limit of generic proxies tested against an archive, not of remote sensing as such.

9 · Screening product: 67 zones (first pass)



67 zones · 556 km² 23.5 % of the domain — screening products, not validated targets.

Redrawn from the first-pass run (21 Aug 2026), before layer screening; to be re-derived from the screened layers. Zone parts on the cover reflect surface signals (sediment type, geomorphology), not mineralization; the cover was not evaluated.

RESULTS

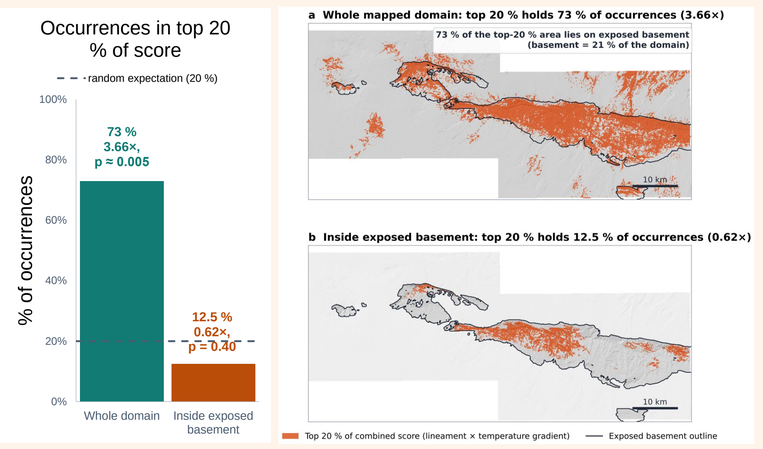
10 · Main result: most of the signal is exposure

The layers find the exposed basement, not the deposits: 3.66× over the whole area → 0.62× inside the basement.

What we tested: best combination — lineament density × four-season Landsat-8 TIRS temperature gradient (lateral, not geothermal); share of occurrences in its top 20 % of area.

What we found: whole domain 73 % of occurrences (3.66×, p ≈ 0.005); inside exposed basement 12.5 % (0.62×, p = 0.40) — below the 20 % expected by chance.

Why it matters: on the map the top-20 % area follows the basement (73 % of it lies on basement, which is 21 % of the domain): the signal is mostly exposure.



Same run as the abstract: 100 m grid; significance from the toroidal-shift null (500 shifts). Occurrence locations not shown (closed data).

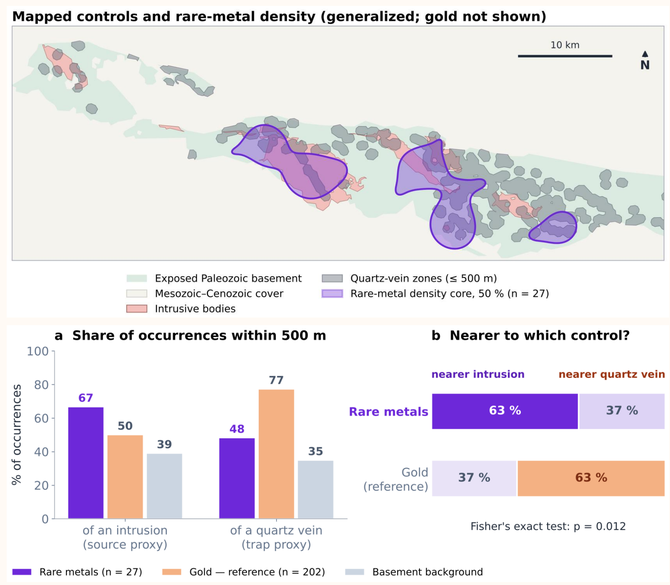
11 · Inside the basement: system-specific proxies work

Rare metals → intrusions (source) · gold, reference → quartz veins (trap)

Approach: the proxy test was first checked on gold, where data are sufficient (trap proxy: distance to quartz veins), and then applied in the same way to rare metals (source proxy: distance to intrusions; panel 6).

What we found: rare metals are nearer intrusions in 63 % of cases, while gold is nearer veins in 63 % (Fisher p = 0.012; 0.009 in a second dataset). Quartz-vein proximity: 2.75×, p = 0.040.

Why it matters: where generic layers failed (panel 10), process-specific proxies separated the systems; the opposite preference of gold rules out an exposure artefact. Exploratory (27 occurrences) — the basis for REE/Li system models and field checks.



Exposed basement, 100 m grid; map generalized; rare metals as a density core (50 %, σ = 1.5 km), gold not shown (closed data).

12 · Answer, limitations, next steps

Answer: much of the regional signal reflects basement exposure (3.66× → 0.62×), not prospectivity within the basement. Inside the basement, each system follows its own geological control — intrusions for rare metals, quartz veins for the gold reference.

What the mineral-systems approach gave: it turned *preservation* (basement exposure) into an explicit baseline, without which 3.66× would read as success. Generic proxies failed inside the basement, while controls derived from each system's processes — the trap for orogenic gold, the magmatic source for rare metals — separated the two groups. Because proxies come from processes, not from training data, 27 rare-metal occurrences were enough to *test* a hypothesis, though far too few to *train* a data-driven model — a typical situation for under-explored critical metals.

Limitations

- Retrospective test on archive occurrences; exploration history shapes the record.
- Many layers tested and selected on the same occurrences → p-values are nominal; small rare-metal sample → within-basement results exploratory.
- Covered ground (~79 % of the area) not evaluated.
- Only standard index products were tested; targeted spectral methods (unmixing, REE/Li signatures, field spectra) were not.

Next steps

- Critical-metal (REE, Li) system models with commodity-specific spectral proxies, tested in the same framework; gold as the reference system.
- Quantify value beyond exposure with independent / spatially separated validation and field checks.
- Interpret the temperature-gradient signal geologically.

References

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Pianowski L.S., Holm-Denoma C.S., Smith S.M., Taylor C.D. (2024) U–Pb zircon geochronology and whole-rock geochemistry data for Auminzatau and Kuldjuktau mountains, Uzbekistan. USGS data release, doi:10.5066/P9ILRITQ.

Data availability: the occurrence catalogue is proprietary (Ministry of Mining Industry and Geology of Uzbekistan); gold locations are not shown, rare metals only as a generalized density core. Remote-sensing data are open access.